RR-F-221/3A
June 2, 1976
SUPERSEDING
Fed. Spec. RR-F-221/3
September 7, 1973

FEDERAL SPECIFICATION

FENCE POSTS AND ACCESSORIES

(DETAIL SPECIFICATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers specific requirements for fence posts and accessories.

1.2 Classification. Posts and accessories shall be of the following types and styles, as specified (see 6.1 and figure 2).

Type I - Posts.
Style 1 - T section.
2 - U section.
3 - Angle section.

Type II - Accessories.
Style 4 - Stays.
5 - Fasteners.
6 - Gates.

Type III - Special.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

FF-N-105 - Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
QQ-S-781 - Strapping, Steel, Flat and Seals.
RR-F-221/GEN - Fencing, Wire, Fence Posts and Accessories.
PPP-B-26 - Bag, Plastic, (General Purpose).
PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-D-723 - Drums, Fiber.
PPP-S-760 - Strapping, Nonmetallic (and Connectors).

Federal Standards.

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals, Test Methods.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 5660

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-packaging, Methods of.
MIL-T-704 - Treatment and Painting of Material.

Military Standards.

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

A 90 - Methods of Test for Weight of Zinc-Coated (Galvanized) Iron and Steel Articles.
A 120 - Specification for Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A 123 - Specification for Zinc (Hot-Galvanized) Coatings and Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars, and Strip.
A 153 - Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A 641 - Specification for Zinc-Coated (Galvanized) Carbon Steel Wire.
E 8 - Tension Testing of Metallic Materials.
E 10 - Brinell Hardness of Metallic Materials.
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 376 - Coating Thickness by Magnetic Field or Eddy Current (Electromagnetic) Test Methods, Rec. Practice for Measuring.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Type I, posts. Posts shall be hot-rolled steel or re-rolled rail steel.

3.1.2 Type II, accessories.

3.1.2.1 Styles 4 and 5, stays and fasteners. Stays and wire fasteners shall be formed from galvanized steel wire.

3.1.2.2 Style 6, gates. Fabric for gates shall be woven steel wire. Gates frames shall be tubular steel, formed steel or rolled steel. Fittings including hinges and latches for gates shall be cast iron or malleable iron, or formed from steel strip or slit sheet.

3.1.3 Types I and II, chemical composition. Unless otherwise specified (see 6.1), chemical composition of posts and accessories shall be at the manufacturers option.

3.2 Type I, posts.

3.2.1 Styles 1, 2, and 3.

3.2.1.1 Yield, tensile strengths, and hardness. Unless otherwise specified (see 6.1), posts shall have a minimum yield strength of 50,000 p.s.i. and a minimum tensile strength of 80,000 p.s.i. or have either a minimum hardness of 156 Brinell or 83 Rockwell B (see 4.4.1).

3.2.1.2 Weight of posts. Unless otherwise specified (see 6.1), the weight of the as-rolled posts, prior to painting or galvanizing, shall be as follows:

Style	Weight of posts (Pounds per foot)
1	1.33
2	1.12 or 1.33
3	4.10 upright members
3	3.19 bracing members

Tolerance on weight of as-rolled shall be plus or minus 5 percent.

3.2.1.3 Dimensions.

3.2.1.3.1 Cross-sectional dimensions. Unless otherwise specified (see 6.1), the nominal cross-sectional dimensions for styles 1, 2, and 3 posts shall be as follows:

Style	Cross-section (Inches)	Thickness (Inch)
1	1-3/8 wide, 1-3/8 depth	1/8
2 1.12 lb/ft	2 by 1-1/8	3/32
2 1.33 lb/ft	2 by 1-1/4	3/32
3 Upright	2-1/2 by 2-1/2	1/4
3 Bracing	2 by 2	1/4

3.2.1.3.2 Length of posts. Unless otherwise specified (see 6.1), posts shall be of the following lengths with a tolerance of minus 1 inch, plus 2 inches.

Style 1 - 5, 5-1/2, 6, 6-1/2, 7, 8, or 10 feet.
2 - 5, 5-1/2, 6, 6-1/2, 7, 7-1/2, or 8 feet.
3 - 7, 8 or 9 feet, uprights and braces.

3.2.1.4 Surface finish. All posts shall be zinc-coated, painted, or unpainted as specified (see 6.1).

3.2.1.4.1 Zinc coat. When specified (see 6.1), posts shall be galvanized in accordance with ASTM A 123 or A 153, as applicable.

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3.2.1.4.2 Paint coat. Cleaning treatment and painting shall be in accordance with MIL-T-704, type A for two coats or type C for one coat. Coating shall be weather resisting paint or enamel baked-on or air dried. Paint color shall be as specified (see 6.1). When specified (see 6.1), paint color shall be in accordance with Fed. Std. No. 595. Unless otherwise specified (see 6.1), thickness of paint shall be at the manufacturer's option.

3.2.1.4.3 Reflective tops. When specified (see 6.1), reflective tops shall be furnished. Unless otherwise specified (see 6.1), the color and length of the reflective portion shall be at the manufacturer's option.

3.2.2 Styles 1 and 2.

3.2.2.1 Style 1, T-section posts. T-section posts shall have studs placed and constructed to engage a substantial number of fence line wires in proper position. The distance between studs shall not be greater than 3 inches.

3.2.2.1.2 Style 2, U-section posts. U-section posts shall have corrugations, punched tabs, notches, holes, knobs, or studs, as specified (see 6.1), and so placed and constructed as to engage a substantial number of fence line wires in proper position.

3.2.2.2 Anchors. Unless otherwise specified (see 6.1), each style 1 and 2 post shall have a steel anchor plate 18 square inches minimum in area and weighing not less than 0.67 pounds, securely fastened to the post. Design and placement of anchor shall be as shown in figure 3.

3.2.3 Style 3, angle-section posts and braces. One angle brace shall be furnished with each end or gate post, and two angle braces shall be furnished with each intermediate brace or corner post, complete with zinc-coated hardware for assembly. When specified (see 6.1), hardware including bolts and nuts shall be galvanized (zinc-coated) in accordance with ASTM A 153 or A 123.

3.3 Type II, accessories.

3.3.1 Style 4, stays. Stays shall be 9-1/2 gage, zinc-coated twisted wire designed for a screw-on type installation. The wire shall be zinc-coated in accordance with ASTM A 641, class 1 or class 3, as specified (see 6.1). Length of stays shall be 24 to 60 inches in 2 inch increments, as specified (see 6.1).

3.3.2 Style 5, fasteners. Wire fasteners shall be formed from zinc-coated wire, 12-1/2 gage or coarser, as specified (see 6.1). The wire shall be zinc-coated in accordance with ASTM A 641, class 1 or class 3, as specified (see 6.1).

3.3.2.1 T-section posts. Unless otherwise specified (see 6.1), fasteners shall be designed for use with T-section posts, size 1-3/8 inches by 1-3/8 inches by 1/8 inch thick. Each end of the fastener shall be bent and twisted up or to the side. The fastener shall be easily installable with ordinary pliers.

3.3.3 Style 6, gates. The style of woven wire and the design including dimensions of gates and gate accessories shall be as specified (see 6.1). Unless otherwise specified (see 6.1), fittings which include latches shall be provided for gates and gate posts. Gate frames and hardware shall be zinc-coated in accordance with ASTM A 120 or A 153, as applicable.

3.4 Type III, special. This type of post and accessories is defined as any type which does not fall within the other type classifications of this specification, but the materials, workmanship and quality shall meet the applicable provisions of this specification.

3.5 Workmanship. Zinc-coated posts and accessories shall be free of bruised or scaled coating, blisters and uncoated areas. Painted posts and accessories shall be free from cracks, peeling, chipped paint, and uncoated areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. See GEN part of this specification.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.1), a lot shall consist of all posts and accessories of the same type and style offered for delivery at the same time.

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4.2.2 Visual and dimensional examination, end item. Sampling for these characteristics shall be level S-2 for posts and level S-4 for accessories in accordance with MIL-STD-105.

4.2.3 Sampling for tests. One post or gate, complete with gate hardware, shall be selected from each lot for testing the weight of zinc-coating.

4.3 Examination.

4.3.1 End item. Posts and accessories selected in accordance with 4.2.2 shall be examined for defects listed in table I. The Acceptable Quality Levels (AQL's), in accordance with MIL-STD-105, shall be 2.5 percent defective for major defects and 6.5 percent for total defects.

TABLE I. Classification of defects, end item

Examine	Defects	Major	Minor
Posts and accessories	Type and style not as specified.	X	
	Damage or defects affecting function or serviceability.	X	
	Damage or defects not affecting function or serviceability.		X
Posts	Length not as specified or not within tolerance.	X	
	Not painted or zinc-coated as specified.	X	
	Anchor plate not securely fastened to posts.		X
	Corrugations, punched tabs, notches, holes , knobs, or studs for U posts not as specified.	X	
	Angle braces, assembly holes or hardware missing.	X	
	Cross-sectional dimensions, when required, not as specified.	X	
	Unpainted or uncoated areas.	X	
	Cracked, peeling, or chipped paint.		X
Accessories	Stays, wire fasteners not zinc-coated.	X	
	Wire fasteners not designed for the posts specified.	X	
	Gates, style of woven wire or accessories not as specified.	X	
	Gate hardware not as specified.	X	
	Gate hardware missing.		X

4.3.2 Length of posts. Measure length of posts using a rigid or tape rule graduated in 1/16 inch increments.

4.3.3 Weight of posts. After measuring length of posts, weigh them and calculate weight per foot (see 3.2.1.2).

4.3.4 Inspection of preparation for delivery. The packaging, packing, and marking of fence posts and accessories shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

4.4 Test methods.

4.4.1 Tensile and hardness properties, type I posts. Determine yield and tensile strength in accordance with ASTM E 8 or for hardness either ASTM E 10 or ASTM E 18 (see 3.2.1.1).

4.4.2 Weight of zinc coating. Determine weight of zinc coating in accordance with ASTM A 90, A 120, A 123 or 153, whichever is applicable. The magnetic test in accordance with ASTM E 376 for nondestructive measurements of the thickness of zinc-coatings is permitted for gates and gate hardware of irregular shape, such as gate latches and fittings.

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4.5 Rejection and retest. Unless otherwise specified (see 6.1), rejection and retest shall be in accordance with the general section of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Posts. Posts shall be packaged and packed in accordance with level A requirements of MIL-STD-163.

5.1.1.2 Stays. One-hundred stays of the same size shall be bundled and tied with a minimum of 3 ties of wire reinforcing tape spaced equidistant along the length of the bundle.

5.1.1.3 Fasteners. Fifty fasteners of the same style shall be secured in a fabric or plastic bag. Forty bags of fasteners shall be packaged in a close-fitting burlap bag conforming to PPP-B-35, type I, or plastic bag conforming to PPP-B-26.

5.1.1.4 Gates. Each gate with fasteners, fittings and latches shall be packaged in accordance with the requirements of MIL-P-116, method III.

5.1.2 Level B.

5.1.2.1 Posts. Five posts of the same size and style shall be bundled with the anchor plate ends alternated. Each bundle of posts shall be strapped with two flat steel straps size 5/8 by 0.020 inch minimum conforming to QQ-S-781, type I, finish A or B, grade 2. Each band shall be securely tensioned and positioned with a band at each end of the bundle. Fifty bundles of posts of the same style and size shall be packed into a lift of 250 post. Each lift shall be laid on two nominal 4 inch by 4 inch wood skids the same length as the width of the skid. The wood skids shall be spaced equidistant from the ends of the lift, with a minimum of 4 feet between the two skids. Anchor plates shall be positioned so they will not project outward from the bundle. Each lift shall be banded with two bands of flat strapping 1-1/4 inches by 0.035 inch minimum, conforming to QQ-S-781 over the 4 inch skids, tensioned and held in place with a nail driven into the end of the skid.

5.1.2.2 Stays. Stays shall be packaged as specified in 5.1.1.2. Ten bundles, 1000 stays, shall be secured into a lift and tied with a minimum of three pieces of flat or round wire strapping, or non-metallic strapping conforming to PPP-S-760.

5.1.2.3 Fasteners. Fasteners shall be packaged as specified in 5.1.1.3. Two bags of fasteners shall be packed in a fiber drum conforming to PPP-D-723, type II, grade A. When palletization is required (see 6.1), four fiber drums of fasteners shall be strapped on a pallet conforming to figure 1. Strapping used for securing drums to the pallet shall conform to QQ-S-781, finish B, grade 2.

5.1.2.4 Gates. Each gate shall be shipped loose, with fasteners, fittings and latches securely fastened to the gate with wire.

5.1.3 Level C. Fence posts and accessories shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification and the National Freight Classification Rules.

5.2 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.1).

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type and style (see 1.2).
(c) Chemical composition, when required (see 3.1.3).
(d) Size of lot, if different from 4.2.1.
(e) Rejection and retests (see 4.5).
(f) Selection of applicable level of packaging and packing required, and when required, palletization for fasteners (see 5.1 and 5.1.2.3).
(g) Marking of shipment (see 5.2).

Posts

(a) Yield and tensile strengths or hardness (see 3.2.1.1).
(b) Weight (see 3.2.1.2).
(c) Dimensions (see 3.2.1.3.1 and 3.2.1.3.2).
(d) Surface finish (see 3.2.1.4).
(e) Zinc coated posts (see 3.2.1.4.1).
(f) Painted posts (see 3.2.1.4.2).
(g) Reflective tops (see 3.2.1.4.3).
(h) Style 2, protrusions and holes (see 3.2.2.1.2).
(i) Anchors (see 3.2.2.2).
(j) Style 3, zinc coat on hardware with posts (see 3.2.3).

Accessories

(a) Style 4 stays, length and zinc coating (see 3.3.1).
(b) Style 5 fasteners, wire size and zinc coating (see 3.3.2).
(c) Fasteners for different size T-section post (see 3.3.2.1).
(d) Style 6 gates (see 3.3.3).

6.2 Fence post styles. The fence post styles illustrated in figure 2 represent typical cross sections and features for fastening fence wire to posts. The fence post styles shown are illustrative only and are not intended to preclude similar designs of fastening features otherwise conforming to the requirements of this specification.

MILITARY INTERESTS:

Custodians:

Navy - YD
Air Force - 84

User interest:

Navy - MC, CG
Army - ME, CE

Military Coordinating Activity:

Navy - YD

PREPARING ACTIVITY. GSA - FSS

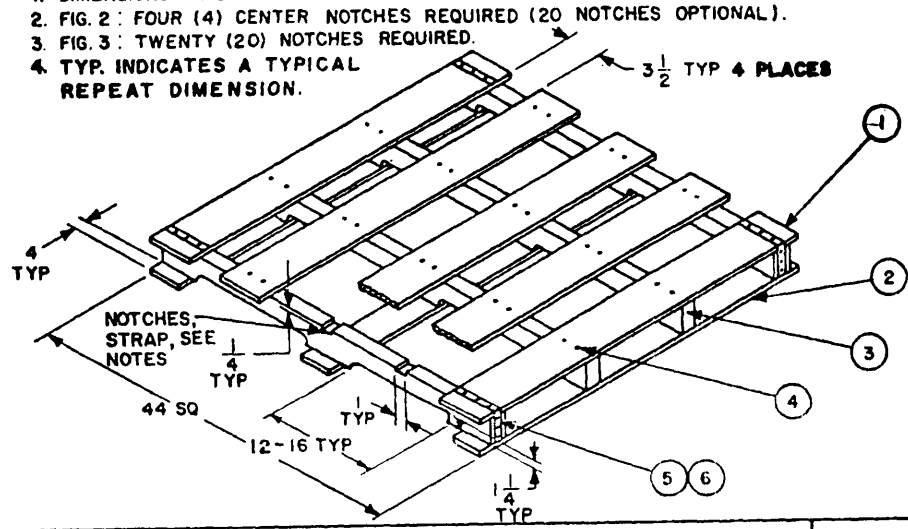
CIVIL AGENCY COORDINATING ACTIVITIES:

Commerce - NBS
HEW - FEC
Interior - BPA
USDA - AFS

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NOTES:
1. DIMENSIONS ARE INCHES.
2. FIG. 2 : FOUR (4) CENTER NOTCHES REQUIRED (20 NOTCHES OPTIONAL).
3. FIG. 3 : TWENTY (20) NOTCHES REQUIRED.
4. TYP. INDICATES A TYPICAL REPEAT DIMENSION.

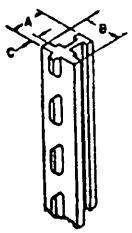


FIND NO.	PART OR IDENT NO.	QTY REQD	DESCRIPTION	SPECIFICATION
6	TYPE II, STYLE 4	24	NAILS, BOX, STD, FLAT HD, DIAMOND PT, STL, CEMENT COATED, 8d X 2-3/8 LG	FF-N-105
5	TYPE II, Finish A	4	STRAPPING, FLAT, NAIL ON, STL, 11/4 NOM SIZE X .035 NOM THK X 9 7/8 LG	QQ-S-781
4	TYPE II, STYLE 18	72	NAILS, PALLET, FLAT HD, DIAMOND PT, STL, ZN COATED, 9 GUAGE X 3 LG	FF-N-105
3	GROUP II, III OR IV, CL 2	4	WOOD, STRUCTURAL, 2 X 4 X 44 LG, CUT-OUT 1 1/4 DP X 16 LG EXCEPTIONS: SAME AS ITEM 1	MIL-STD-731
2	GROUP II, III OR IV, CL 2	3	WOOD, STRUCTURAL, 1 X 4 X 44 LG EXCEPTIONS: SAME AS ITEM 1	MIL-STD-731
1	GROUP II, III OR IV, CL 2	5	WOOD, STRUCTURAL, 1 X 6 X 44 LG EXCEPTIONS: HEMLOCK UNACCEPTABLE, DOUGLAS-FIR MUST BE COAST-TYPE	MIL-STD-731

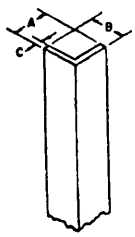
LIST OF MATERIAL

FIGURE 1. Pallet construction

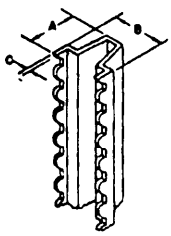
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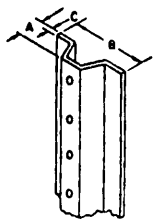
Style 1



Style 3



Style 2



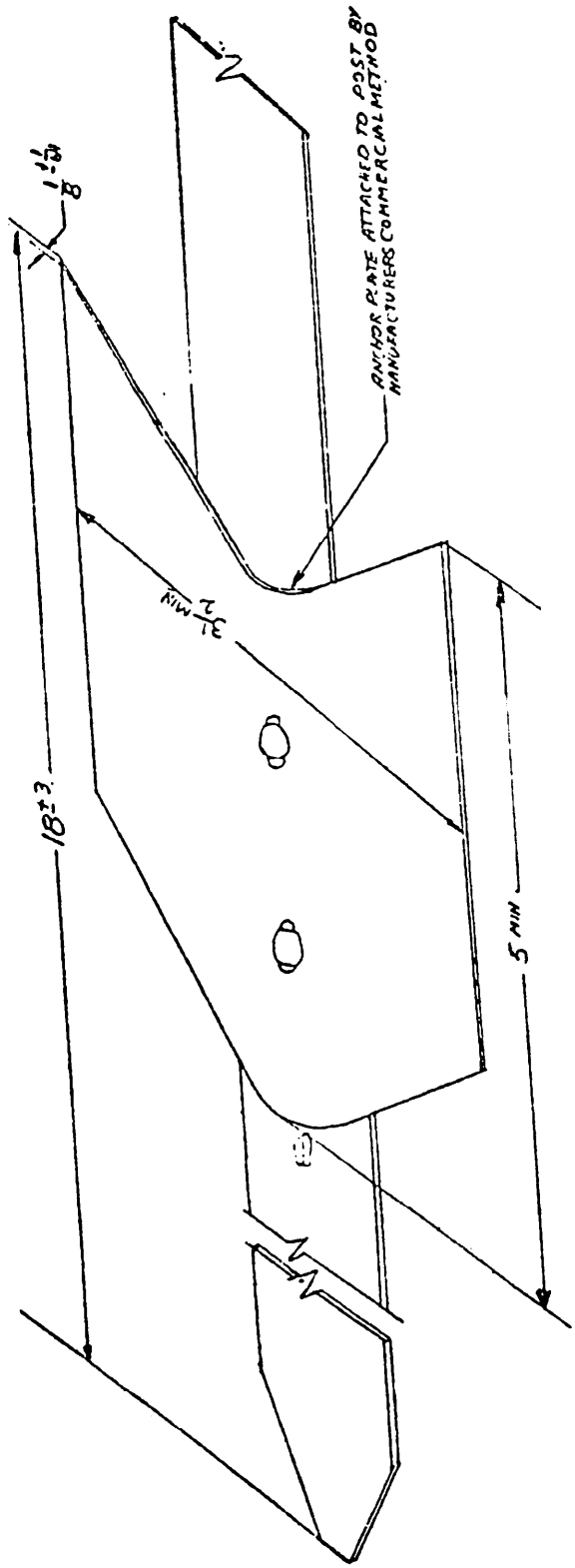
Style 2

Legend

A-Depth
B-Width of face
C-Thickness of material

FIGURE 2. Fence post styles

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Anchor design and placement

FIGURE 3.

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